

The Efficacy of Legislative Intent as a Macroeconomic Predictor: A Compositional NLP Analysis of U.S. Appropriations Bills

By Jack Trepp & Abhinav Bonagiri

1. Abstract

1.1. Summary

This study investigates the relationship of governmental policy, mainly budget appropriations, with the macroeconomic performance of the country through analyzing the text composition of the bills. To narrow the focus, the U.S. will be the only country analyzed for this study. By leveraging Legiscan datasets, we analyzed the composition of all U.S. spending legislation passed from 2009 onwards. We utilized Zero-Shot Classification, a Natural Language Processing technique that classifies data that hasn't been seen in model training using semantic relationships, to categorize appropriations into distinct sectors, allowing for the mapping of shifts in spending priorities over time. These categorical proportions were cross-referenced against U.S. economic indicators to analyze changes in economic performance between categories to determine if they can serve as predictors of economic health.

Our analysis reveals that, under normal economic conditions, there is a negligible correlation between the thematic distribution of bills and economic health. However, under conditions of severe macroeconomic volatility, such as significant market contractions or pandemics, we found that there is a divergence in spending allocation. Specifically, a divergence that followed the COVID-19 pandemic, where the correlation between fiscal allocation and economic performance showed a marked increase. These results suggest that while the composition of government spending has no predictable composition in periods of stability, its association with the economy becomes pronounced during periods of economic instability.

1.2. Research Questions

- **How Often Are Certain Appropriations Bills Used?**

Identifying trends in the implementation of different policy types to help understand the impact of those policies, which can be done by understanding what typically occurs before a certain type of policy is implemented, such as the state of the economy.

- **How Do Different Types of Bills Impact the Economy?**

Policies are usually implemented in a systematic manner, and we can identify these patterns by identifying similar attributes across different policies. By breaking policies down into these attributes, we can determine which parts of an economic policy have the largest impact, as well as trends and relationships between these parts, such as one part only being effective when combined with another. We can also determine if certain policies are only effective when the economy is acting or trending a certain way.

- **How Long Does It Take for a Policy to Take Effect?**

Some policies may not result in an immediate impact on the economy, such as being delayed or never occurring, which may hurt the economy instead of helping it. By looking for trends in the implementation of a policy and the delayed impact on the economy, we can identify how much the policy lags behind the current health of the economy.

2. Results

2.1. Workflow

For our research questions, we are concerned with two groups of data: economic indicators such as GDP and Consumer Confidence, and the bills we analyzed. The processing for the economic data is simple: combining split datasets, replacing missing values with interpolated averages, and rotating the data as needed. The processing for bill data is much more difficult, involving webscraping and one-shot encoding classifications.

The list of bills we obtained is first filtered to only include appropriations bills that were passed. The URLs contained in the dataset are then scraped in order to obtain the entirety of the bill's text. This includes traversing several links within the page in order to reach and download a copy of the bill.

This text copy is then saved and used as a text to be classified using either sentence-transformers/all-MiniLM-L6-v2 model or Moritz Laurer's deberta-v3-large-zeroshot-v2.0 model. It is classified into several categories, which we choose to represent the majority of bill types, including Military, Scientific, and Educational spending.

Once the economic data is prepared and the categorization of all the appropriations bills is complete, we create several plots comparing the bill counts, the proportions of bills, and the economic data in order to search out potential trends within the data.

2.2. Sources of Bias

- **Data only included US Bills:**

Using bills from just one country may introduce sources of bias through variables such as population size, the country's general culture, and so on. This means those lurking variables are not controlled, and their influence may cause errors when applying this data to other countries.

- **Time Period:**

The bills we analyzed only dated back to 2009, limiting the historical scope of our findings. The more the economic health is being tested and brought back to normal, the

Additionally, this time period contained the COVID-19 pandemic, which impacted the global economy, potentially introducing errors into our findings. (The data benefits from volatility showcasing how governments react to poor economic health)

- **Bill Number vs Budget:**

Our research focused on the number of bills rather than simply the government budget, but both variables are tied together. The variance in budget per bill can cause misleading conclusions if conclusions are made about spending rather than the bill number. Additionally, a year with a large number of appropriations bills cutting certain types of funding would show an increase in that proportion, whereas the proportion of that category in monetary terms would be decreasing, which is important to keep in mind when making conclusions with our data.

- **Budgets in other Bills:**

Sometimes governments attempt to slip legislation into budget bills and budgets into legislation bills. This can potentially confuse the classification algorithm through unrelated points being covered in the appropriations bills, whilst omitting data through budgets in other bills excluded from our dataset.

- **Correlation vs Causation:**

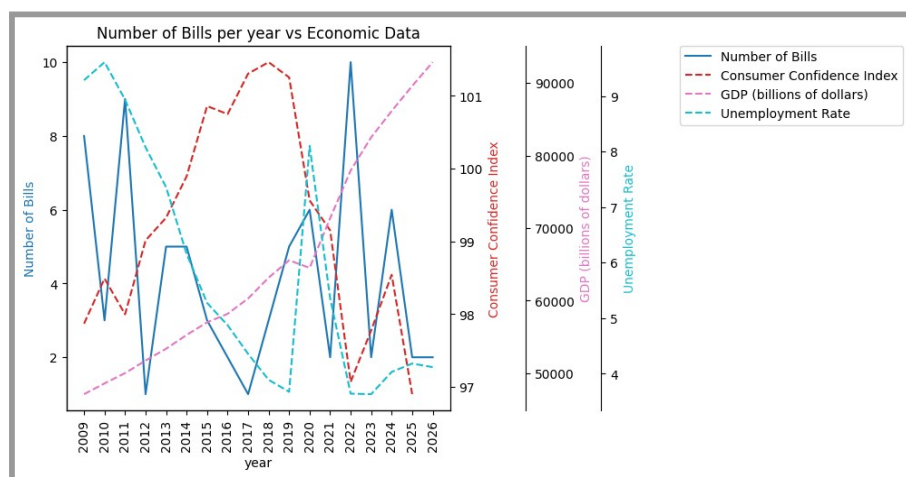
Since our findings are a study and not an experiment, we are unable to claim that any trends we notice are the work of bills on the economy rather than the impact of the economy on policy or some third variable causing both changes.

- **No Distinction Between the Size of Bills:**

Supplemental Bills are treated on the same level as big appropriations bills, while their actual impact may not reflect this equality.

2.3. Plots

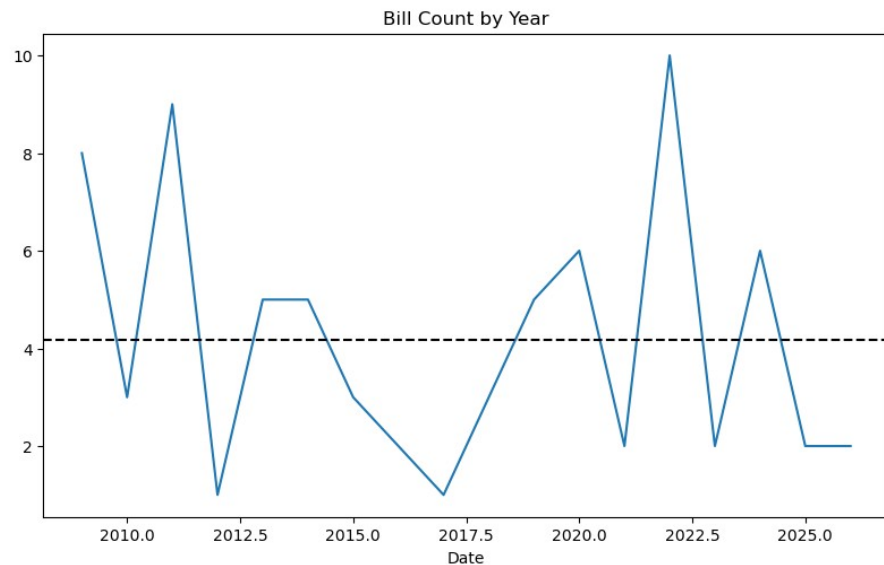
2.3.1. Number of Bills



(Fig 1)

In Fig 1, we can see that there appears to be little to no correlation between the number of bills and economic indicators. Despite the unemployment rate appearing to follow the number of bills sometimes, the correlation is still weak, with the signs of their trends often being random (if the number of bills is trending up, we can't easily predict if unemployment will trend up or down)

We can also see that Consumer Confidence drops and unemployment spikes in 2019 at the start of the COVID pandemic, while GDP only takes a small hit, implying that GDP is not the most accurate indicator of economic health, at least on a large, multi-year scale.

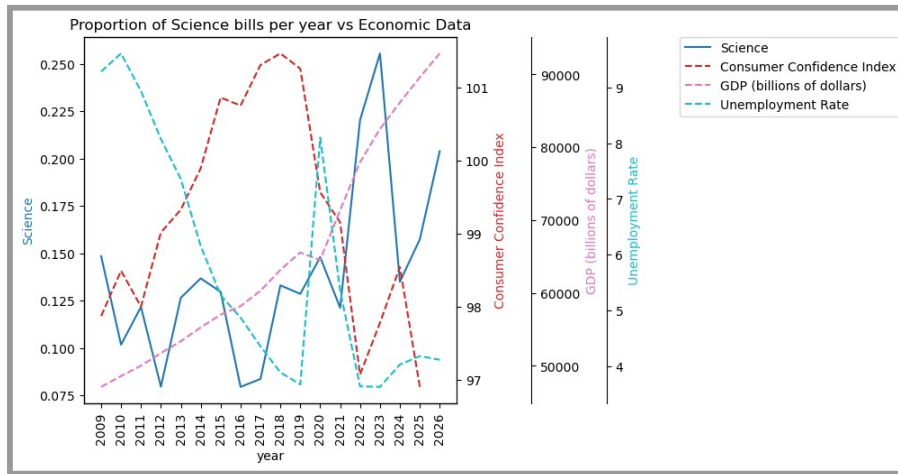


(Fig 2)

In Fig 2, we can more clearly see that over the last 18 years, there is no real trend in the number of appropriations bills passed, with the amounts varying between one in 2017 and ten in 2022.

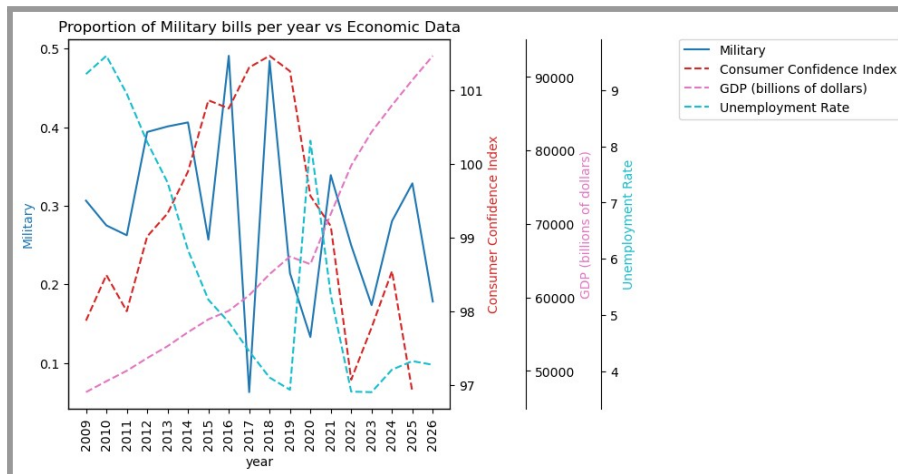
However, the average number of bills is $4.1\overline{66}$ with a standard deviation of 2.749... bills. This tells us that the distribution of the number of appropriations bills is skewed right, meaning years with large amounts of passed appropriations bills are much rarer than years with only a few appropriations bills.

2.3.2. Proportion of Bills



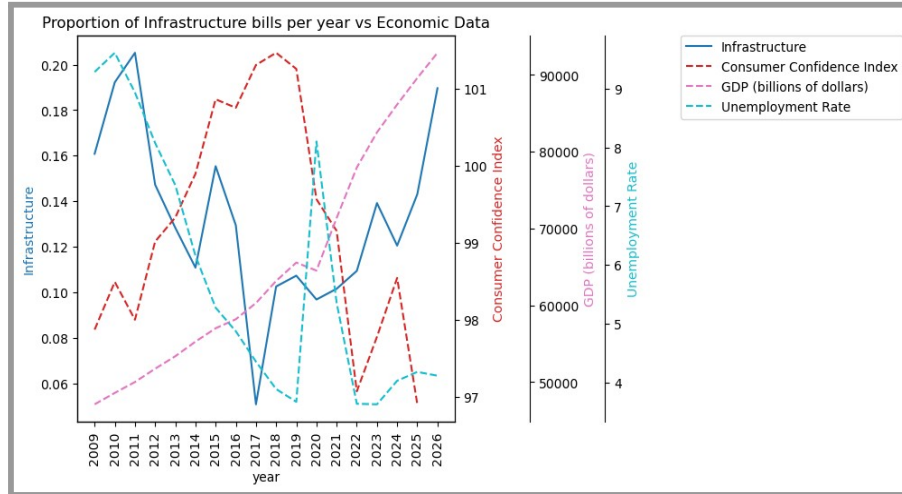
(Fig 3)

In Fig 3, we can see the proportion of bills relating to scientific spending appears not to correlate with Unemployment Rates or Consumer Confidence; however, it does appear to follow the general trend of GDP, increasing over time. It also appears to be the opposite of the changes in consumer confidence following 2021.



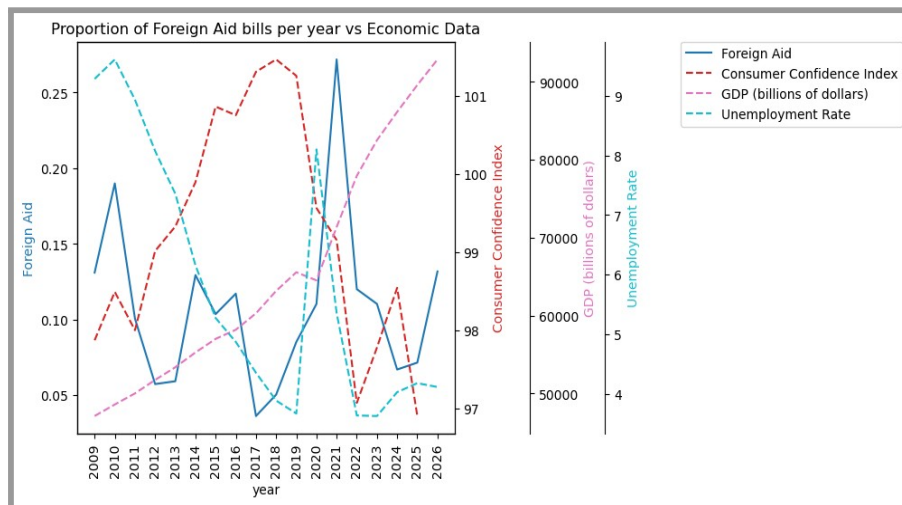
(Fig 4)

In Fig 4, we can see the number of bills relating to military spending appears to slowly decrease over time, but with large dips and spikes throughout, and very little correlation with any of our economic indicators.



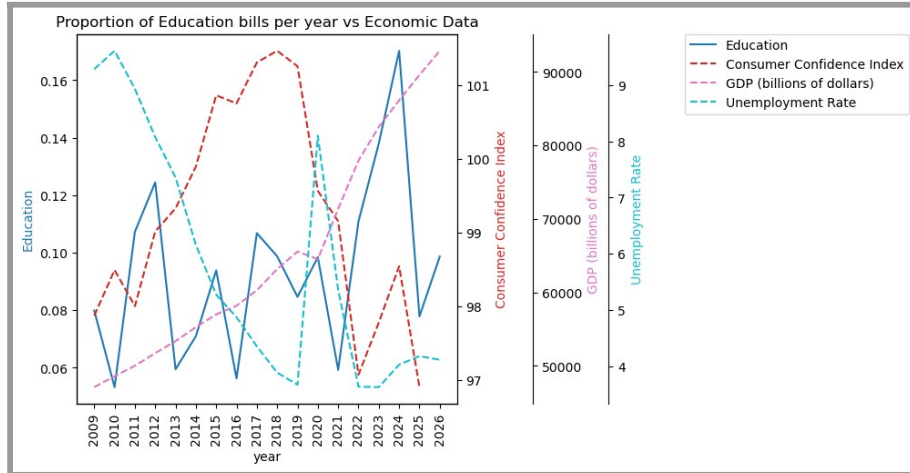
(Fig 5)

Fig 5, the graph for the proportion of bills relating to infrastructure has a weird pattern, trending down along with unemployment until around 2017, before trending up with GDP. Other than those two disconnected possible correlations, there seems to be no correlation with the economic indicators



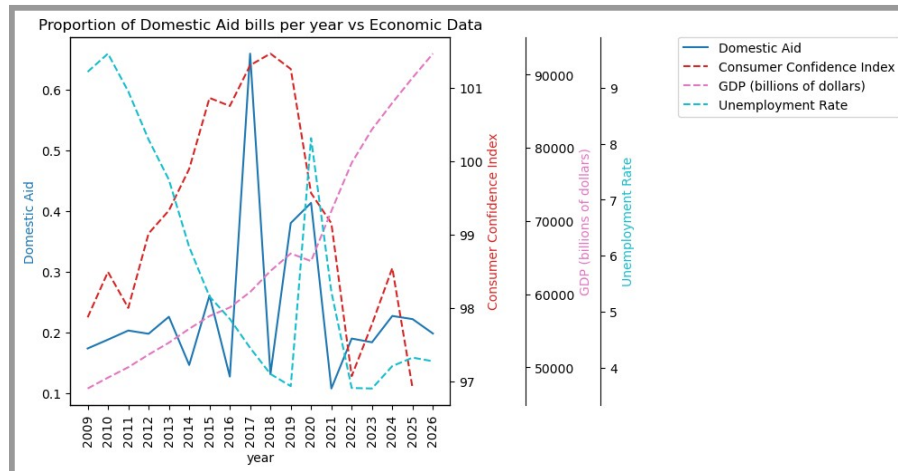
(Fig 6)

In Fig 6, foreign aid spikes shortly after 2019, almost perfectly mirroring the 2019-2020 spike in unemployment rates, although about a year later. Other than this spike, there appears to be no correlation or trends in the data.



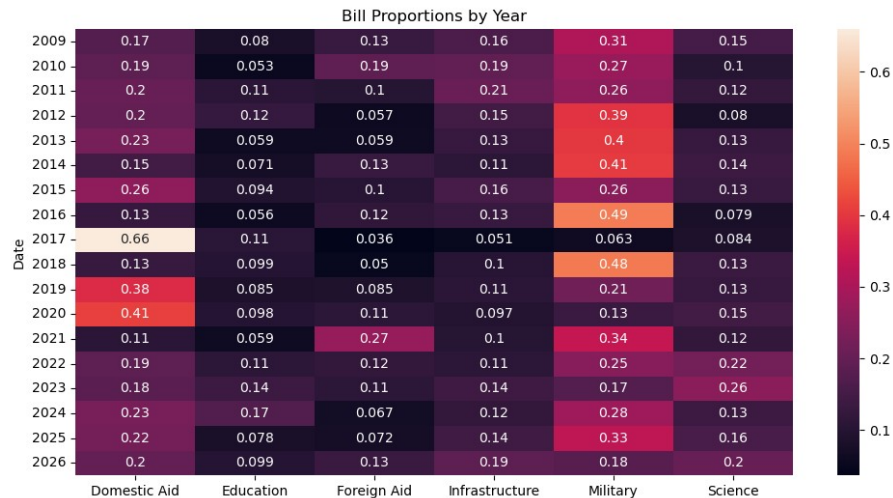
(Fig 7)

In Fig 7, education appears to be slowly increasing in proportion over time, somewhat following GDP. It also appears to reflect consumer confidence in recent years (2022 onwards), as both appear to increase and decrease at the same time. However, there appears to be no correlation with unemployment or consumer confidence prior to 2022.



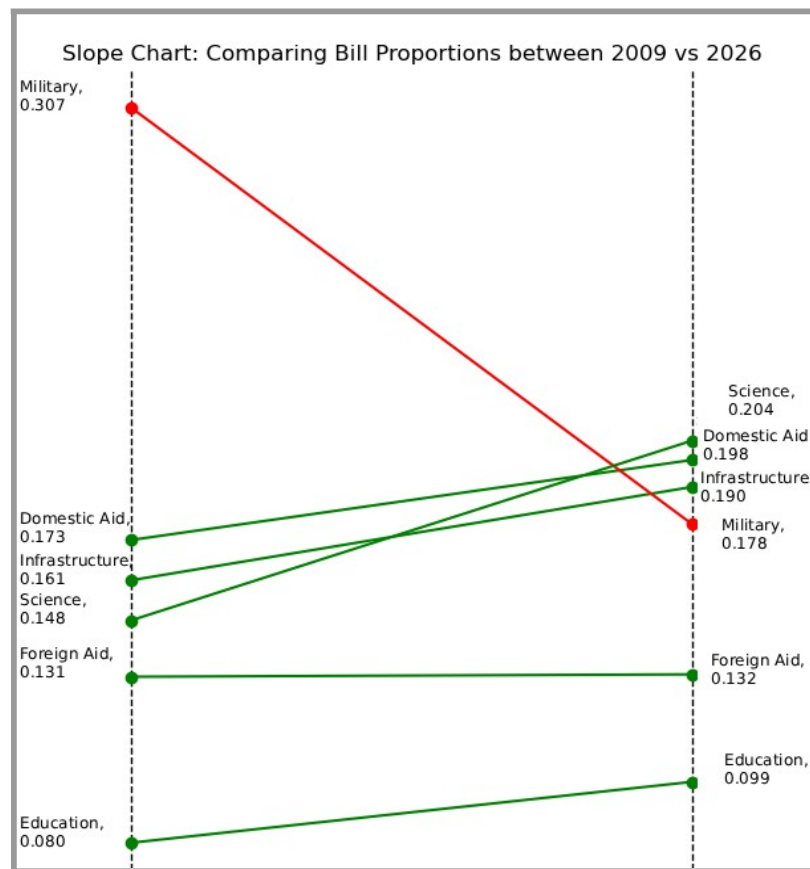
(Fig 8)

Fig 8 shows that domestic aid seems to be a relatively low proportion of all appropriation bills, although the data suggests a massive increase in 2017. After 2021, it seems to follow changes in consumer confidence and unemployment rates — especially the direction of those changes a year later — although the correlation is minor and apparently nonexistent before that point.



(Fig 9)

Fig 9 helps visualize the proportions of our different classifications of appropriations bills over time. It showcases the massive spike in domestic aid in 2017, and the proportion of military spending seems consistently high, whilst education remains low.



(Fig 10)

Comparing just 2009 to 2026 in fig 10, we can see that all categories have a higher proportion in 2026 except for military spending. We can also see that scientific spending had a large increase, and foreign aid had a very minor increase.

3. Discussion

The impact of the COVID-19 pandemic can be easily seen in the economic data, with GDP and consumer confidence dipping in 2019 and unemployment increasing. Many of the other oddities in the graph can likely be explained through a connection with COVID-19. For example, the spike in foreign aid, as seen in Fig 6, could potentially be the US increasing foreign aid following the pandemic. Perhaps the correlations between domestic aid (Fig 8) and education (Fig 7) with consumer confidence and unemployment rates after 2019 may be due to changing culture, politics, and economic situations following the pandemic, with people becoming more hopeful in the economy as the situation improves through education and aid.

An interesting data point is the correlation between consumer confidence and unemployment rates. Prior to 2019, they seemed inversely related, with consumer confidence increasing as unemployment decreased. This seems to make sense; as it becomes harder to get jobs, people would naturally become more pessimistic in their views of the economy. However, this pattern changes from being inversely related to being mostly random but with a more positive correlation. This seems to imply that people view large unemployment numbers as a positive reflection of the economy, but may instead just be waves of relief and pessimism at the economic situation following COVID.

It's also likely that the increase in appropriation bills relating to science may be in part due to the Artemis program and in part a miscategorization with bills lowering scientific funding and cutting grants that were seen at the start of Trump's second term.

4. Conclusion

4.1. Findings

For each type of appropriations bill, we found that the correlation between the proportion and economy, whether that relationship is positive or negative, typically became stronger during and following the COVID-19 pandemic, suggesting that the proportions and economy have a much higher association during these times of economic crisis.

Specifically, bills targeting education appeared to closely mirror trends in the economic situation after 2019, while bills targeting domestic aid funding seemed to predate trends in the economic situation, with consumer confidence typically increasing within a year or two of an increase in domestic spending. Meanwhile, it appears that increased foreign aid spending tends to follow improvements in the economic situation, suggesting governments are more likely to spend money on improving the situation in other countries once the situation in their own has improved.

4.2. Limitations

- **US Only:**

Because our data only came from the US, it should only be used to make predictions about the US. Making predictions about other countries from just this data would lead to potentially inaccurate results.

- **Altering Number of Bills:**

Our findings should not be used as justification for increasing or decreasing the number of bills by splitting up appropriations among many bills or combining what would be many supplementary bills into a single, large bill. Our data only works if it remains representative of the scales of bills. By altering these numbers, our data would likely no longer hold.

Furthermore, since our data can not confirm causation, just correlation, it is possible that there would not even be an economic impact from purposefully altering the number of bills.

Likewise, if the number of bills or their size drastically changes, our findings would likely become outdated and, as such, shouldn't be used in this scenario.

- **Longer Time Frequencies:**

Our analysis was done on timespans of one year, with an entire year of economic changes and legislation compressed into one data point. This limits our ability to draw conclusions about short term (less one year) associations, and may compress delays of less than one year into appearing as an instant change.

5. Further Research

- **Extending Backwards:**

Our data on bills only reaches back to 2009. By expanding the timespan covered by our appropriations dataset, we would improve our ability to find trends, including long-term trends in the data. Including additional economic disasters, such as the Great Depression, could help confirm the conclusion that the correlation between bills and the economy increases during and after times of crisis.

- **Extending to Other Countries:**

Our bills were only sourced from the US, limiting the scope of our research. Having access to appropriations data from other nations would allow us to better determine if potential trends are actual trends or just coincidental, while also expanding the conclusions we can draw to the entire world.

- **Budgets not Just Bills:**

Expanding our analysis to include budgets laid out in the bills rather than just categorization would allow us to draw more accurate conclusions on how government spending impacts the economy, and also allow us to search out additional trends in the relationship between the number of bills and the actual budget.

- **Expanding Legislative Data and Other Governmental Actions**

Laws of every nation limit businesses and individuals from certain spending, the creation of certain goods, or promote them as well. Things such as tariffs, tax cuts,

subsidies, and interest rates can change how a business operates. These governmental actions can impact how an economy produces goods and services and, therefore, its economic health. But it is worth noting that adding complexity can introduce concerns such as

- **Machine Learning to Analyze Trends:**

Training a machine learning algorithm to predict the compositions of bills based on our findings could help uncover long-term trends and associations in the data, such as some bills having a large impact on the economy several years later that we wouldn't be able to notice on our own.

- **Causation or Just Correlation?**

Running an experiment, likely simulated, could potentially help determine if the correlations we found are just that, correlations, or if there is causation one way or the other. This could also determine which way the causation is--how much the proportion affects the economy and how much the economy affects the proportion of bills--if said causation exists.

- **Why Did Science Appear to Have a Negative Correlation with the Economy?**

Our data seemed to show a negative correlation between bills targeting scientific spending and the economy, which seems odd. Further research could find an answer or disprove a correlation. Was this result just a coincidence? Was it a large number of bills targeted at cutting scientific spending? Or do bills targeting science actually correlate to worse views of the economy? Or is it instead a reflection of the economy, not an impactor of it, with science being cut in times of need? Further research could help us understand the actual relationship between scientific spending and the economy, which could reflect on people's views of scientific research.

- **How Does the Political Climate Connect to Our Findings?**

Our data just focused on the United States, but the associations between the economy and the proportion of bills may differ in countries with different political climates or even between terms of the US as democrats and republics gain or lose power. By studying other countries and longer time spans within the US, we could identify trends between not just bills and the economy, but bills, the political landscape of the country at the time that determine the legislative intent, and the economy, adding additional depth and accuracy to our analysis.

6. Challenge Goals

6.1. Webscraping

The project required web scraping documents from Legiscan, a website that stores legislative data, which can be accessed through an API. While the API is sufficient for use with 30000 requests/month, we decided to use webscraping as a challenging goal, unlike using the API, which requires less effort to set up. Indeed, there were many issues with Legiscan scraping, as the PDF was hidden under subtabs and links, which were heavily protected by the PDF viewer. To avoid this issue, there was a table that included the links to PDF files from congress themselves, which avoids this issue. Normal libraries, such as Beautiful Soup 4, were

not able to load JavaScript-loaded HTML tables, where the PDF URL was located. To overcome this challenge, SeleniumBase was used to run a virtual headless browser that masquerades as a “human” user to scrape data. By doing this, I can overcome the extreme security of the Cloudflare bot verification system due to the inability of the bots to press buttons.

6.2. NLP Processing Documents

To analyze our appropriations, we utilized zero-shot classification to analyze and categorize the spending into labels: ["Education", "Military", "Infrastructure", "Science", "Domestic Aid", "Foreign Aid"]. These labels are used to identify where the budget was being allocated rather than getting numbers from the documents. Many NLP processing workflows can be used, such as cleaning data and extracting data. But, minimal cleaning of data is needed as advanced NLP models tend to use the context of grammar, punctuation, and prior context to help base relationships and predictions. For the purpose of

Why only use Zero-Shot Classification?

There are issues if we had used other NLP models, including the fact that not all numbers represent the budget that will be spent. If we were to include it, this would misattribute the spending or increase it dramatically due to a number without knowing its context. There are many variables to account for when counting the budget using an appropriation bill. To simplify this process, zero-shot classification is used as it goes beyond just sentence embeddings and classifies relationships with categories through semantic intent using Natural Language Inference.

CPU - Sentence Embeddings

This is a different alternative to zero-shot classification for researchers who don't have a GPU at hand. Sentence embedding is similar to Zero-Shot Classification, but finds semantic relationships by comparing sentences using numerical vectors. While Zero-Shot Classification is more accurate due to its inference capabilities, sentence embeddings replicate it at a lower level and are efficient in terms of computing power.

7. References

- [Consumer Confidence Index \(Data Source\)](#)
- [World Economic Outlook \(Data Source\)](#)
- [Legiscan Bills \(Data Source\)](#)
- [IDP Website \(Outline, Code\)](#)
- [Python Gallery \(Slope Graph Code\)](#)
- [Geeks for Geeks \(Code Help\)](#)
- [Python Stack Overflow \(Code Help\)](#)
- [Matplotlib Documentation \(Code Help\)](#)